

Bullet Hells and the Information Problem

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I've been spending a bit of time with *Returnal* lately, and it's been a game that I am finding incredibly disappointing, and yet still feel the urge to play. There is quite a bit that I think has been done well in the game, and what it does well is what makes me want to revisit it. But the core elements of the gameplay just keep feeling...off.

To begin, *Returnal* is an over-the-shoulder shooter that relies on two core elements to set it apart: roguelike elements and bullet hell combat. Each of these elements have problems, based on how they are incorporated into the game. I want to visit each one in turn, but for this essay I'm going to focus on the latter.

The idea of a "bullet hell" is pretty simple to grasp. You control some character or vehicle, and as enemies would spawn on the screen they would throw some kind of attack at you which would need to be dodged. So far, this is pretty standard. What made bullet hells unique was the sheer number of attacks that would be thrown at you at once. Usually keeping track of one or two enemy attacks is easy, but as you throw in more and more the player's focus begins to shift, so that rather than trying to kill enemies before they attack, the game becomes more about weaving between enemy shots while trying to launch attacks of your own. Bullet hells are generally more defensive than offensive in nature: it's more important to not get hit than it is to land your own blows.

There are plenty of great examples of bullet hells of lots of varieties. The classics include games such as *Gradius* and *Ikaruga* – 2D shooters in which the player controls a spaceship. Roguelikes are certainly no

stranger to the concept: *Binding of Isaac* and *Enter the Gungeon* both rely heavily on bullet hell gameplay, particularly in their boss battles.

And there are games where bullet hells aren't so good. *NieR* comes to mind as a game that frequently throws in bullet hell mechanics, but rarely do those mechanics serve an interesting purpose. They are an obstacle to the player, but the obstacle is so easily dismissed that it loses its danger.

So there are plenty of examples of bullet hells to look at. What I intend to do here is dig into what makes bullet hells work, and show how *Returnal* falls short in capturing those components. There are a variety of changes that could have been made to make the overall system mesh well, but as-is the game appears built around a system that is fundamentally unfair.

Bullet Points

Bullet hell games rely on two key facets of gameplay skill: reaction and situational awareness. In order for the game to actually use its bullet hell system properly, players need some way to be able to escape danger, which requires avoiding these bullets. But while avoiding a bullet that is about to hit you is one thing – this might test reaction itself – bullet hells throw in a problem regarding the future. There's not just a single bullet about to hit you, but other bullets coming at different times and different angles. So you need to be aware of all of those other bullets and where they're heading.

In a normal game, you are dodging attacks one by one. This bullet is about to hit you – dodge it. This next bullet is about to hit you – dodge it. Your actions in how you avoid the first bullet do not impact how you react to the second.

But in a bullet hell, that cascading effect matters. It's possible to dodge the wrong way, so that you can avoid the first bullet, but get hit by the second. Or avoid the first and second, but get hit by a third. Bullet hells require that you not just react to immediate threats, but plan ahead to avoid future threats at the same time.

To make all of this work, you need to feed a player the right information. Obviously, dodging all of these attacks requires some level of skill, so it's not as though getting hit is a fault of the game's design. But it *can be* if the player does not have some way of knowing how to avoid attacks.

This can happen in several ways. The attacks themselves could come out of nowhere, preventing the player from having any ability to register what is happening before they get hit. The attacks could be too hard to avoid, meaning that even when you try to dodge, you still get hit. And sometimes there can be so many attacks that it can be impossible to dodge everything.

Certain methods for how we normally experience bullet hells illustrate how to combat these problems. Most bullet hells – particularly successful ones – are 2D. All attacks generally exist within the player's view, so it's almost impossible to be blindsided. Many bullet hells make attacks a sort of slow or moderate speed, so that players can easily assess what the attacks are and how to dodge them effectively. And in addition, many bullet hells try to balance the mixture of enemies to prevent there being too much at once.

While the concept of throwing stuff at the player may sound easy enough, actually balancing everything to make the trial fair is a difficult task.

A Real Bullet Hell

Returnal is designed to be hard. In fact, the game opens up by warning the player that it will be hard. There is definitely value in doing so, and there is nothing wrong with making a hard game.

However, a difficult game is not inherently "good." There is a line that can be crossed when making a game difficult that ultimately disrupts the game's attraction. If a game veers in the direction of being unfair, if it frustrates players to a degree that the game is no longer worth playing, then something is probably wrong. A game can be designed to be unfair (and sometimes players will enjoy those games merely for the extreme

difficulty), but unfairness can also be a sign of bad design. And we cannot point to "difficulty" as a savior.

So how does *Returnal* stack up against these three pitfalls of bullet hell design?

Well, when we look at the first component, *Returnal*'s over-the-shoulder camera means that it's possible for players to be hit from behind or from the side by attacks they couldn't see. The game tries to account for this problem by adding a small indicator on the character to show when attacks are coming from offscreen. That should solve the problem, right?

Well...no. And to explain, allow me to go on a short digression about perspective. Video games of all types can often throw a lot of information at a player. More, really, than the player is able to process. To get around this, players will generally focus on particular parts of the action and particular areas of the screen, concentrating on elements that demand the most attention. Other parts of the screen can still receive some attention, but usually in a much lesser form: they'll get brief glances.

For example, think about things like life bars or ammo counters. These things are usually tucked into the corners of the screen. Part of the reason for that is to prevent them from being a distraction. It would be hard to fight enemies if they're obscured by your own HUD. But just as important is that there's not a problem with them being tucked away like that. You don't *need* to be constantly aware of how much health you have. Instead, you can just glance now and then to check, and then plan accordingly. When you're normally playing, the health bar or your ammo counter is just a blur of information that only comes into focus when you consciously think about it, and then goes back into the blur once you return to the gameplay.

Returnal's perspective relies on the area surrounding the character. The real estate of concentration starts in the middle and then radiates outward. But importantly, the real estate doesn't really extend to the player character. There's generally no information about the character that

can be usefully gathered by observing her.

With one exception: the indicator for when bullets are coming at you from offscreen. This indicator is placed on the character, near the bottom of the screen. In fact, it's not just in a bad place of screen real estate, but in the furthest part of that bad place. It's the periphery of the peripheral.

This poses a problem because this kind of indicator isn't something you want players to be "occasionally checking," as they do with health or currency. You want their focus drawn to it. But because the player is also dealing with bullets *in front of them*, their attention is divided looking up and down at the same time. Dividing attention like this means something has to give, and what will give is the indicator. And that means players are going to be hit from offscreen.

How about speed? *Returnal* has some enemy variety, with two or three different attacks. The most common attacks are bullets fired in particular patterns – a classic bullet hell staple. But added to this can be attacks that track the player, such as lasers and missiles.

Attacks that home in on a player can be tricky. Since the point is to make them difficult to dodge, if they're not done right then they tend go from being "tough" to "impossible." While *Returnal* doesn't throw any individual attacks that are impossible to dodge, it is possible to face enemies in later areas that throw multiple homing attacks at you. Attacks that, on their own, you can dodge. But which fired at a somewhat constant pace means any attempt to dodge will be successful, only to leave you vulnerable to another attack. Because the game relies on three forms of dodging – strafing, running, and a dash – there are multiple defensive options that can interfere with each other. The dash needs a couple seconds to recharge, but it is also the most effective means to avoid attacks. Comparatively, strafing works for a lot of attacks, but is probably the least effective method to employ, almost never being able to avoid homing attacks. Running, being in the middle, can sometimes avoid homing attacks, but at the cost of needing to move quickly in a straight line – which can mean running blindly (not a great idea) or redirecting your

camera to give you a clear line of sight as you run (which means attacks might hit you from offscreen as you try to get away).

This problem also speaks to the game's balancing. Most of the time, the enemy variety and their attacks line up the way you would expect a bullet hell game should. There are a lot of bullets to dodge, but as long as you understand the pattern, you should be fine. But once the game starts throwing in large numbers of enemies at you, and especially when those enemies start using homing attacks, the balance goes off kilter.

Rebuilding *Returnal*

Importantly, these three issues I just described aren't actually fatal to *Returnal*. If we were dealing just with these elements, the game would be tough, but not necessarily unfair.

The final problem, the straw that breaks the camel's back, is the fact that the game is also *punishing*. Getting hit can take a large chunk of health, which means being able to avoid attacks is absolutely necessary. Once this punishment gets added in, these problems rise to the level of unfairness.

Does this mean that *Returnal* as a concept is fundamentally unworkable? No. There are still ways we can make it work, but something ultimately has to give. Some value of the game's design needs to be sacrificed.

For example, let's say it was of primary importance that the game be punishing. Okay, that's fine. Punishing does not mean unfair. It's only unfair when the game is punishing in a manner that the player is unable – or almost unable – to avoid.

So rebalancing the bullet hell would be top priority. One thing that would need to be done is reworking the enemy AI so that it focuses on moving around players and attacking them from the front. This might seem odd, but it helps to give players a sense of orientation about the dangers coming at them. Obviously, there would still be attacks that come from offscreen, because the game couldn't just make bullets disappear when

the player isn't looking at them (it could, but that would end up being rather silly and make the game too easy), and so some reworking of the threat indicator would be necessary to give it a better place of importance on the screen, somewhere that players could more easily reference it while also keeping track of the bullets onscreen. But even with a better threat indicator, the problem of offscreen hits is still going to be a problem, because there is only so much information that such an indicator could provide, compared to actually being able to see the enemy's attacks for yourself.

I'll note that this rebalancing would allow the game to remain an over-the-shoulder shooter. Given many of the examples I've used for bullet hells earlier, and the criticism I laid down regarding being hit from offscreen, it might seem like the only solution would be to remove the 3D aspect entirely, and that only 2D bullet hells can work. I don't think that's necessarily true. [There's at least plenty of exploration of the combination that first needs to be done](#) before we can confidently arrive at that conclusion.

But let's say instead that you were willing to give up the punishing aspect of the gameplay. In which case, while it might be annoying to be hit from offscreen or be unable to dodge attacks, the game can be rebalanced by giving players more health and more healing options. Rather than the player dying in 5 or 6 hits, the player might have 15 or 20. The game would be less difficult, but in return the unfairness would be made a bit more fair. The player does not need to feel frustrated by the game itself, and any failure to progress can be better chalked up to the player's failure than to the game's design.

These are just a couple potential solutions, and ultimately the ideal solution depends on what the primary focus of the game is. It's important to keep in mind that conflict between different elements of a game is not just possible but common, and when that conflict arises it becomes imperative to settle on what the core value of the game is. Whether it's difficulty, or replayability, or storytelling, or whatever, it's necessary to

prioritize that core value and figure out how to make it work properly.

Concluding Remarks

The appeal of bullet hells is that they are difficult, and yet difficulty is not merely about killing the player. There is a fine line between a game being genuinely tough and a game being unfair, and few games successfully walk that line, and sometimes even the games that do occasionally trip into unfairness.

What's key to understanding difficulty and fairness, though, relies on understanding the medium the game is ultimately using to convey that difficulty. Whether it's through exploration, or puzzles, or a lack of options, or ambiguity, or just throwing enemies at the player, these different methods all require different approaches. What works to make one difficult game fun and interesting will not work for another game.

So in looking at bullet hells, it's important to think about why bullet hells work in the first place, and then think backwards. If a player's success ultimately relies on information, then you need to think about how to give the player as much (useful) information as possible in a relatively short span of time. Anything that blocks information – from the physical space of the screen to the placement of threat indicators to the sheer amount of enemies and attacks to process – all serve to undermine the core gameplay mechanic, and thus serve to make the game unfair.